

5.6 Alternative processes that can be used to manufacture components and systems to different scales of production	Application, advantages and disadvantages, of the following processes, scales of production and techniques when manufacturing products, in order to be able to discriminate between them and select appropriately for use.
	5.6.1 Processes: - a photo etching - b PCB population – PCB drilling and soldering.
	5.6.2 Scales of production: - a one-off prototyping (breadboard) - b batch - c mass production - d continuous.

SYSTEMS

Key idea	What students need to learn:
	5.6.3 Techniques for quantity production: - a pick and place technology - b surface-mount technology (SMT) - c quality control - d marking-out methods (use of reference points, lines and surfaces) - e templates - f patterns - g sub-assembly - h working within tolerance - i efficient cutting to minimise waste.